Supporting Information

PhytoScope for portable local queries of plant genome data

Supplementary Methods

The software and evidence snapshot is commit `159ae0f79bd3681afc6c914db282c515190a3888` at https://github.com/omixcore/phytoscope. The code requires Java 21. Source accessions, versions, checksums, gene/transcript gold records, and the public expression matrix are supplied with this report. Full genomes, annotation archives and raw reads are fetched from their public repositories rather than duplicated in the submission package.

The three additional RefSeq assemblies are GCF_029865625.1, GCF_009805555.1 and GCF_006381635.1. Official annotation checksums and accession-matched FASTA identify the tested releases. The eight structural inventories independently specify recognized feature types, explicit parents, sequence identifiers, strands and 1-based inclusive coordinates; unsupported feature types are reported separately. Five coding loci per additional species produce six ricebean, 14 Andrographis and five Hibiscus coding-transcript gold records, with independent genomic-span, spliced-transcript and CDS SHA-256 digests.

Reproduction commands

Run in the pinned full repository:

```
python tools/fetch_vigna_reference.py --directory target/non-model/vigna-umbellata
python tools/fetch_additional_nonmodel_refs.py --directory target/additional-nonmodel
python tools/build_nonmodel_sequence_gold.py
./mvnw -B test
```

The ricebean integration test uses the fetched full reference inputs:

```
./mvnw -B -Dtest=NonModelVignaReferenceTest \
  -Dphytoscope.vignaReferenceDir=target/non-model/vigna-umbellata test
```

For the public expression calculation, use Salmon 2.8.0 and the exact RefSeq RNA FASTA/GFF3 named in `public-expression-manifest.json`:

```
python tools/requantify_public_expression.py \
  --rna-fasta target/non-model/vigna-umbellata/GCF_029865625.1_rna.fna \
  --gff target/non-model/vigna-umbellata/GCF_029865625.1_genomic.gff.gz \
  --reads-dir /path/to/fastq --salmon /path/to/salmon \
  --output target/non-model/vigna-umbellata/expression \
  --threads 4 --fetch-reads --temporary-reads
```

The script verifies read byte sizes and MD5 values, exact transcript-to-gene parents, and output hashes. Its mapping contains 54,178 reference transcripts and 36,754 distinct parent genes; the retained Salmon gene-level matrix has 34,914 rows. The manifest key `gene_map_count` counts distinct parent genes, not transcript-map file rows. No tximport length correction or differential-expression inference is applied.

The archived gffutils 0.13 comparison uses 10 Arabidopsis, 24 rice, 20 maize and 40 soybean loci, with three alternating-order fresh-process trials per species and source-derived tuple agreement. Hosted-runner import and SQLite index-build timings are retained in `benchmarks/results/2026-10-18/cross-species-extraction.md`; these operations differ and are not ranked as equivalent workflow timings. Independent promoter, domain, expression-grouping and project-restoration checks use the distinct references and denominators in Table S4.

Table S1. Species and data coverage

Taxon	Assembly / annotation	Data evaluated	Test scope
Arabidopsis thaliana	TAIR10/Araport11; Ensembl Plants release 62	Full annotation, chromosome FASTA, selected expression and protein examples	Full recognized-feature tuple audits; promoter, CDS QC, homology, expression arithmetic, domain mapping; app regressions
Oryza sativa	IRGSP-1.0; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Zea mays	B73 NAM5; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Glycine max	v2.1; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Vigna umbellata	GCF_029865625.1, Vumbellata_1.0; GCF_029865625.1-RS_2026_09	Full GFF3 and GTF; matching chromosome FASTA and RNA FASTA; three public read libraries	Full imports; query/structure; independent selected-sequence gold; Salmon re-quantification and exact ID reconciliation
Andrographis paniculata	GCF_009805555.1; Annotation Release 100	Full GFF3; selected scaffold FASTA	Full GFF3 import/search; selected-locus structure and sequence reconstruction
Hibiscus syriacus	GCF_006381635.1; Annotation Release 100	Full GFF3; selected scaffold FASTA	Full GFF3 import/search; selected-locus structure and sequence reconstruction

Table S2. Full annotation tuple inventories

Inventory	Tuples	Missing	Extra
Arabidopsis bundled annotation	811,827	0	0
Arabidopsis full release 62	791,557	0	0
Maize source excerpt	377	0	0
Maize full annotation	1,161,691	0	0
Rice source excerpt	347	0	0
Rice full annotation	550,921	0	0
Soybean source excerpt	839	0	0
Soybean full annotation	1,324,199	0	0
Total	4,641,758	0	0

Tuple fields include recognized gene, transcript, exon, CDS, and UTR type, explicit parent relation, sequence ID, strand, and exact coordinates. The validator documents ignored or unsupported source feature types separately.

Table S3. Public expression quantification inputs and output

Field	Value
BioProject	PRJNA765494
Genotype / developmental stage	IC426787; 5 days post anthesis
Runs	SRR16122599, SRR16122608, SRR16122607
Assembly / cultivar	GCF_029865625.1 / Vumbellata_1.0 / VRB3
Annotation release	GCF_029865625.1-RS_2026_09
Quantifier	Salmon 2.8.0; selective alignment; k-mer length 31
Reference transcript count	54,178; RefSeq RNA FASTA SHA-256 in public-expression-manifest.json
Distinct GFF3 parent genes / matrix rows	36,754 distinct parent genes / 34,914 matrix rows
Exact annotation ID matches	34,914 of 34,914 matrix IDs; 0 unmatched
Missing / invalid cells	0 / 0
TPM column sums	1,000,000.000034; 999,999.999897; 999,999.999951
Interpretive limit	Genotype differs from reference cultivar; import and identifier compatibility only; no differential-expression inference

Table S4. Additional module validation panels

Function	Independent reference	Observed result	Scope limit
Promoter extraction	Direct 1-based inclusive FASTA slicing and reverse complement; bedtools 2.31.1	8/8 exact sequences, 2,200 bp each, both strands represented	Eight loci on one Arabidopsis chromosome
CDS sequence quality	Independent base-count/frame/start/stop oracle	Four RBCS CDS models (411, 561, 546, 546 nt) matched; no ambiguous bases flagged	Four models from one family
Protein similarity	Test-side Smith–Waterman and BLASTP 2.12.0+	All 16 within-panel alignment metric checks; 12/12 non-self RBCS and 8/8 negative pair classifications agree; 20/20 in the cross-species mature-RBCS panel	Small one-family smoke test; not orthology, sensitivity, or specificity evaluation
Expression grouping	Expression Atlas E-MTAB-7978 source values	Selected means/medians reproduced; measured zeros retained	Single-assay group arithmetic, not differential expression
Domain mapping	Independent Python reconstruction	16 domain/deletion coordinate records across 13 genes matched	Existing annotated models; not a curated multi-species accuracy set
Project persistence	JavaFX regression actions	Save, replace, restore, locus state and outputs verified in CI	Software behavior, not usability study

Supporting Figure Legends

Figure S1. Evaluation coverage. Rows identify seven tested plant species; columns indicate annotation import, search/structure, selected sequence-gold and expression-matrix checks. A dash denotes a task not evaluated in that species.

Figure S2. Structural tuple inventories. Counts of recognized gene, transcript, exon, CDS and UTR relationship tuples in eight source inventories. Bars and exact labels show each count; the summary reports missing and extra records against source-derived references.

Figure S3. Ricebean expression processing. Assigned fragments are shown separately for the three public read libraries. The processing steps identify the fixed RefSeq transcript set, gene-level TPM matrix and identifier reconciliation. Source genotype IC426787 and reference cultivar VRB3 are labeled.

Figure S4. Independent module checks. Panels list task-specific counts for promoter sequences, domain-coordinate records, small protein-similarity classifications and expression identifiers. The protein panel is an RBCS-family smoke test. Separate denominators are retained for each check.